

Chapter 144

Image Processing Order of Operations

The addition of the HDR palette and Color Warper in DaVinci Resolve 17 necessitated an updated diagram of the order of all image processing operations in DaVinci Resolve, from input through output.

A truly detailed breakdown now requires two charts as detailed in this chapter; the first chart details the overall order of operations taking place on every page, while the second zooms into the specific order of operations occurring within each node of the Pre-Clip, Clip, Post-Clip, and Timeline grades that you create on the Color page. Understanding the order of operations presented here will make it easier for you to control the power of the full DaVinci Resolve toolset.

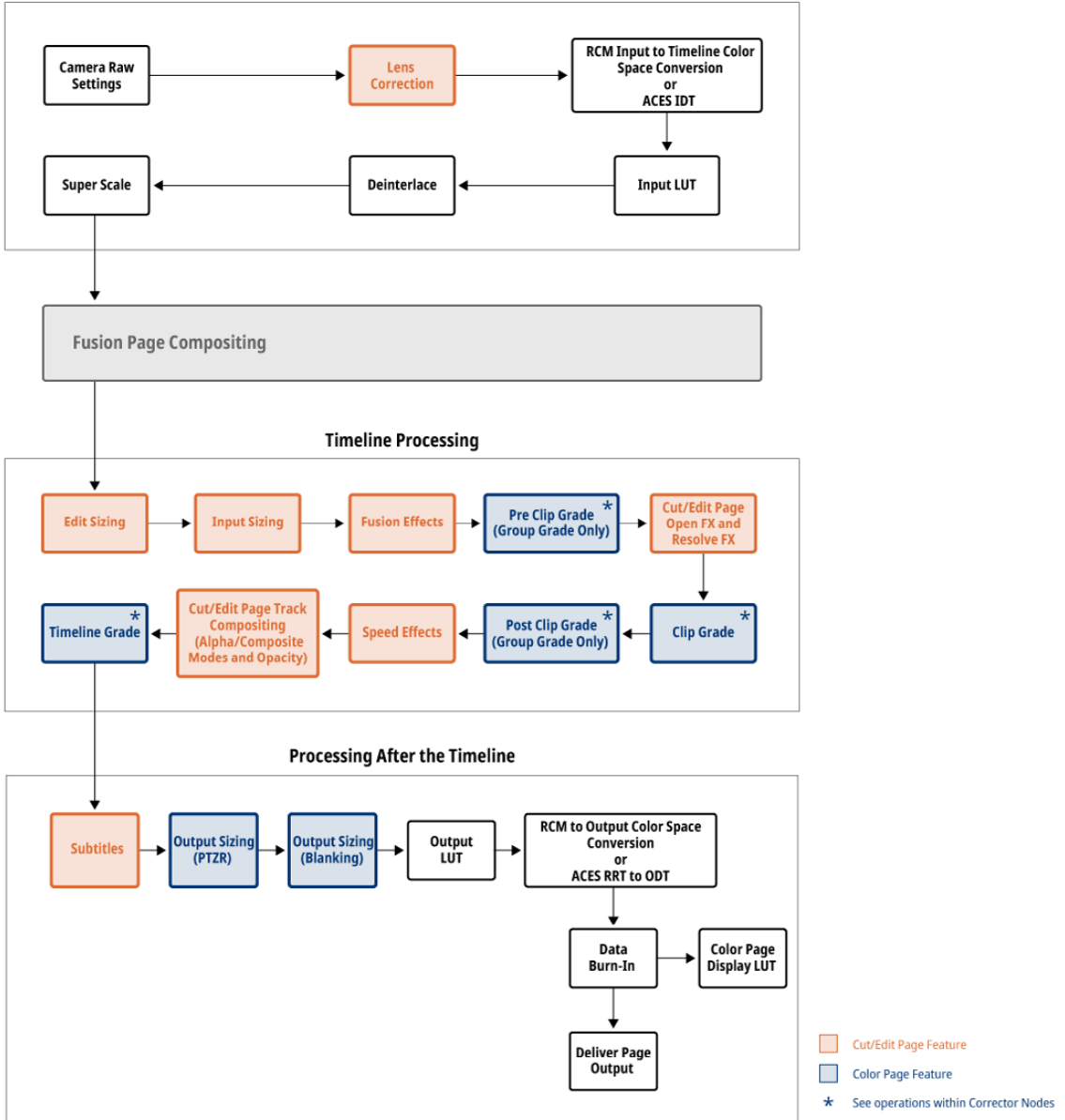
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Overall Image Processing

In the following chart, each image processing operation in DaVinci Resolve is shown in the order in which it's processed. Operations are color-coded to indicate the page on which those controls appear, because in many cases page order does not determine image processing order. Why? Either to make sure each operation is processed as cleanly as possible, or to give the user the maximum control over the image. For both of these reasons, image processing order is carefully considered, and in some cases the best result requires operations on different pages to alternate with one another.

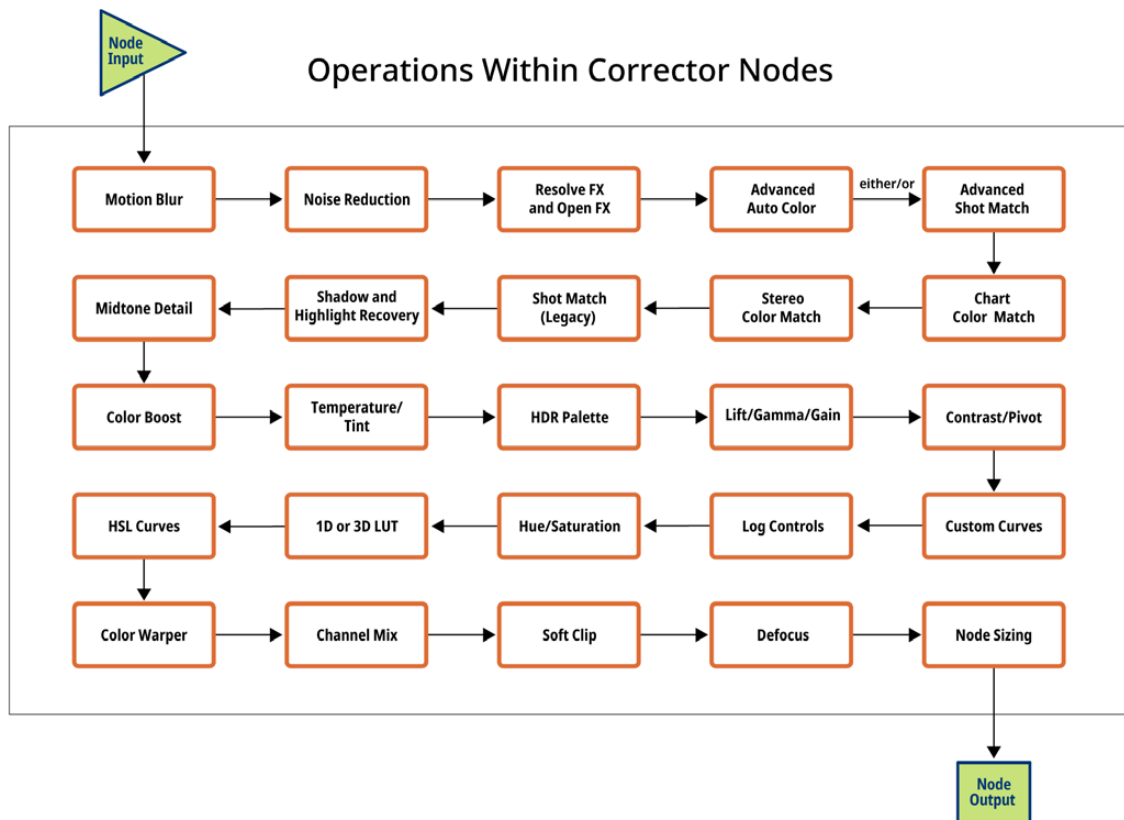
Most of the time, this order of operations is irrelevant to the user who's only interested in the end result. However, if you're trying to achieve something very specific, or if you're wondering why you see a particular result when you use the features of the Cut, Edit, Fusion, and Color pages together all at once, this chart should help make things clear. Finishing artists in particular should find this chart illuminating.



The overall order of image processing operations in DaVinci Resolve, from input through output

Image Processing Within Grades

For Pre-Clip, Clip, Post-Clip, and Timeline grades on the Color page, most Color page operations are available within each Corrector node you add to the Node Editor. Within each node, these operations take place in the following specific order.



The order of image processing operations that takes place within each Corrector node in the Color page

Thanks to the modular operation of the Node Editor, if a pair of image processing operations you need to use isn't in the ideal order for what you're trying to do, you can apply each operation using two different nodes in order to force those operations into the order you want. Similarly, when using Layer, Parallel, or Key mixer nodes, the order of operations can be seen in the visible arrangement of nodes in your node graph.

That said, most colorists tend to spread different operations across multiple nodes, whether they need to or not, for organizational purposes or to "sandbox" certain volatile decisions that might need to be independently revised later. However, nothing's keeping you from using multiple operations within a single node if you prefer a simpler node structure for your grading. Thanks to the flexibility of the Color page, it's your choice.